



Systemic economic impacts of variation in international oil prices: The case of Colombia

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ABSTRACT

This article assesses the systemic economic impacts of oil price fluctuations on the Colombian economy through a calibrated computable general equilibrium (CGE) model for the year 2015. A simulation exercise was conducted, introducing a 1 % positive variation in commodity prices to examine the implicit elasticity of these prices in the country's economy. The results indicate an increase in gross domestic output due to higher investment and government spending. There is also a positive variation in real wages, driving household consumption, along with an increase in the basic prices of the economy. Domestic oil prices exhibited a greater degree of variation than international prices, leading to a loss of the country's external competitiveness. Consequently, economies with low diversification may become vulnerable in volatile price scenarios. The loss of competitiveness in these products can result in an expansion of the trade deficit.

1. Introduction

Historically, developing countries have commodities and primary products as one of the driving forces of their economies. This article aims to measure the systemic economic impacts of variations in international oil prices on the Colombian economy. Through a regional computable general equilibrium model, an exercise will be conducted to simulate a positive 1 % variation in global commodity prices.

According to the International Monetary Fund (IMF),¹ countries are classified into advanced and emerging economies. For such classification, the organization employs various criteria, such as per capita income level, diversification of export baskets, and degree of integration into the global financial system (IMF, 2022).

Colombia² is classified as an emerging economy. Analyzing the country's international integration, a delayed trade openness is noted, becoming evident only after 2002. From the first half of the 21st century, with an increase in exports, there has also been a positive shift in Colombia's gross domestic product (GDP) (World Bank, 2022). In terms

of economic diversification, Colombian exports are based on primary products, especially oil. Between 2015 and 2019, oil was the country's main exported product, representing 38 % of total exports in that time frame. Similarly, manufactured goods represent 21 %, and agricultural products represent 19 % of Colombian exports.³

Given the relevance of oil products in the country's export basket, it is noteworthy to evaluate historically the external participation of this product. The main international oil prices, such as Brent and WTI, are traded on stock exchanges and are heavily influenced by the member countries of the Organization of the Petroleum Exporting Countries (OPEC). Therefore, in the international commodity market, Colombia stands as a price taker and does not exert influence in this regard. Fig. 1 depicts the variation in international oil prices and the export of the commodity by Colombia, highlighting the positive relationship between the two variables.

The assessment of the behavior of the Colombian oil economy is not limited to the observation of international price variables and exports alone. In accordance with Olivera et al. (2011) In 2003, a restructuring

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¹ They comprise those countries included in the Euro Zone, the European Union, or the G7 component countries, located outside the European community, namely, Canada, the United States, Japan, and the United Kingdom.

² For a recent reference for studies that deals with several aspects of the Colombian economy go to Haddad et al. (2023).

³ For more discussion about International Trade in Colombia, Garcia-Garcia et al. (2019) reflect on the obstacles to the growth of foreign trade in goods and the diversification of Colombian exports of goods other than coffee and mineral energy. One indicator that reveals the poor performance of Colombian foreign trade is that the export basket consists mainly of traditional goods, initially coffee, and mining and energy assets in the last years.

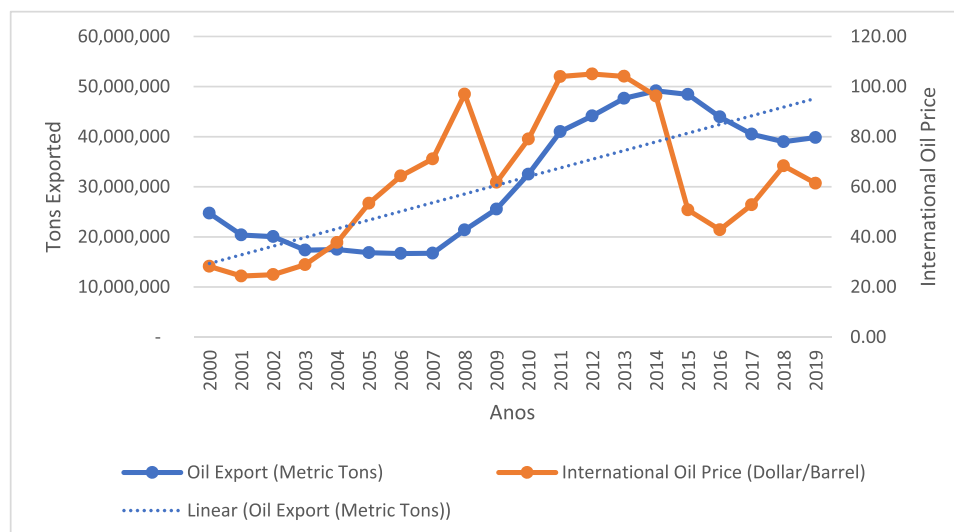


Fig. 1. Exports of oil and derivatives/international oil prices 2000–2019. (US\$).
Source: Prepared by the author based on data from the [Dane \(2000a\),\(2022b\),\(2022c\)](#).

took place in Colombia's oil sector with the creation of the National Hydrocarbons Agency (ANH), which became responsible for the administration and regulation of the country's oil resources. Ecopetrol, the largest oil company in the country, was limited to oil exploration and commercialization, attracting significant inflows of foreign investment. Due to the increase in international commodity prices and the expansion of foreign investment, the country experienced a surge in oil drilling, rising from 10 wells in 2002–112 wells drilled in 2010.

[Olivera et al. \(2011\)](#) also provide an overview of the oil services sector and its evolution over the decades. According to the author, in addition to the expansion in the number of wells drilled, there was a substantial increase in foreign direct investment (FDI) in the oil sector. In 2010, FDI in this sector grew tenfold compared to 2003, representing 42 % of the country's total foreign direct investment.

With the aim of systematically analyzing the impact of variations in global oil prices on the Colombian economy, we define, through hypothetical extraction,⁴ the commodity's production chain in the country, identifying the sectors that could be most affected. Subsequently, from the perspective of computable general equilibrium (CGE), a simulation exercise will be conducted through a positive 1 % variation in global commodity prices. Among the wide range of available CGE models, we choose to use ORANI_G, a regional computable general equilibrium model. The model has been adapted for different countries, including customization for Colombia, developed by [Haddad and Aroca \(2016\)](#) and calibrated based on the Colombian Input–Output Matrix for the year 2015. The regional CGE model adapted to the country allows for a more detailed analysis of the national economy, including more precise intersectoral and systemic relationships, enabling a better assessment of internal impacts following the proposed shock.

This study seeks to contribute to the literature by conducting a simulation exercise that evaluates the long-term systemic impacts of fluctuations in international oil prices on an economy reliant on this commodity. The findings will enable the identification of potential public policies to mitigate the effects of price volatility on the oil production chain, particularly the sectors most affected by price shocks. Additionally, the study maps the production linkages of the oil industry in Colombia, offering a clear understanding of the productive sectors most vulnerable to these fluctuations. Furthermore, the framework developed here can be applied to other oil productive economies in order to compare the degree of dependence upon this commodity using a

systemic analysis.

This article is structured beyond this introduction with a literature review, a methodology and database section describing the methods used and strategies for evaluating the topic from the perspectives of input–output and CGE, followed by the presentation of the results and final considerations.

2. Literature review

As presented in the introductory section, Colombia is a country whose development has been closely linked to the discovery and exploitation of oil, a commodity of significant importance in the country's export portfolio. Thus, to accurately understand the country's participation in the oil market, it is important to assess the historical context and the evolution of oil exploration in Latin America, particularly in Colombia.

According to [Palacios et al. \(2002\)](#), since the 1970s, oil has been one of the main products exported by Venezuela, Mexico, Ecuador, Colombia, Argentina, and Brazil. In 1986, Colombia began exporting products to the international market. Despite the region being a strong oil producer, it is worth noting that there is no self-sufficiency in terms of resources for these countries. Due to this scarce scenario, Latin American countries generally have a trade deficit when the analysis is disaggregated for petroleum-derived products. This is due to the difficulty in refining products by developing economies. Like other commodities, oil exhibits volatility in its price, and economies dependent on the input, as often the case with Colombia, are price takers.⁵ [Braginskii \(2009\)](#) presents the history of international oil prices. According to the author, the formulation of oil prices throughout history has gone through four distinct phases. First, during the 1940s, where prices were determined by a simple single-base system, with production costs being the main determinant; second, a dual-base system was adopted between 1950 and 1970, where it was determined based on a basket of oil products, still with production costs as the main factor; and between 1973 and 1986, prices were determined by an official cost system, taking into account the reference prices of the Organization of the Petroleum Exporting Countries (OPEC). Finally, from 1986 onward, international oil prices were determined by market prices.

The theme of oil dependency and the impacts of variations in commodity prices are also addressed in the literature. [Jiménez \(2010\)](#)

⁴ For more details on the extraction method, see [Miller and Blair \(2022\)](#).

⁵ For more details on oil prices, see [Braginskii \(2009\)](#).

indicated that Colombia began its oil exports late, after 1986, when some mineral reserves were discovered. During the period from 2000 to 2008, oil exports represented between 3 % and 5 % of Colombia's gross domestic product (GDP). The author emphasizes that a significant part of some of the inputs required for the country's oil production originate from imports, and investments are made through foreign direct investment (FDI). Therefore, due to this dependency on the external sector, price volatility can affect production across the country.

The author indicates that commodity prices have direct and indirect impacts on a country's economic growth and other macroeconomic variables, such as exchange rates and tax variables. At the time of Jiménez's publication (2010), oil exploration and exportation were returning significant revenue volumes to the country's economy.

As demonstrated, since the expansion of oil exploration, Colombia has become dependent on this commodity and, consequently, suffers from the effects of price volatility associated with it. This phenomenon is part of a broader pattern observed in resource-dependent economies, where reliance on commodities can, according to some authors, negatively impact a country's development. In economics, the concept of the "Dutch Disease" is widely used to explain the phenomenon where a national economy is adversely affected by the discovery of natural resources, due to rising international commodity prices and the appreciation of the real exchange rate. Furthermore, the literature discusses the "resource curse hypothesis," where natural resources have a long-term negative impact on the economy. Various authors address this topic and present different findings.

In some studies, it is argued that there is no conclusive evidence of a robust resource curse, as demonstrated by Arin and Braunfels (2018), who use a Bayesian model to assess the existence of this phenomenon in 91 oil-exporting countries. Betz et al. (2015) also find no evidence of such a phenomenon in the U.S. coal mining sector. Through a cross-country estimation for 100 countries between 1970 and 2000, Brunnschweiler (2008) points out that there is no evidence of a negative effect on economic growth due to natural resource abundance.

Conversely, Atkinson and Hamilton (2003) present, using a cross-country model, that the resource curse should not be analyzed in isolation but as a result of the mismanagement of available resources. From this perspective, the authors argue that countries where growth has contracted are those where the combination of natural resources, macroeconomic policies, and public spending management has led to low savings rates, resulting in slow growth.

Fleming and Measham (2013) approach the discussion surrounding the resource curse from a different perspective. Focusing on regional development, the authors conclude that the "curse" remains an open debate in economic discourse, and that the presence of natural resources can present both costs and benefits for different countries, especially from a regional perspective, where the spatial location of resources is important and results in distinct socioeconomic impacts. In another study, Fleming et al. (2015) empirically evaluate the regional impact in Australia and conclude through an econometric model that the discovery of natural resources was a "blessing" for local economies near the resource sites. However, negative effects were experienced in other parts of the country.

On a broader scale, Mehrara and Mohaghegh (2011) evaluate the macroeconomic dynamics of oil-exporting countries through an econometric model for the period from 1985 to 2009. Their conclusion is that commodity price fluctuations significantly affect the macroeconomic aggregates of oil-producing and exporting nations.

To analyze this issue in the context of Colombia, several studies where the country is the central focus will be presented. The resource curse, particularly concerning the Colombian experience, is addressed by Perry and Olivera (2010). This study measures the impacts of mineral exploitation from the perspective of regional and local development in Colombia. The findings indicate that oil production positively impacted per capita GDP at the municipal level where natural resource extraction occurred. However, at the state level, a negative effect was observed,

especially in states overly dependent on oil. In general, the authors conclude that mixed effects exist regarding the presence of a resource curse in Colombia. Furthermore, the study emphasizes the importance of institutional quality as a determinant of the impact of resource abundance on growth.

At the regional level, Gaviria et al. (2002) present the impacts of oil discoveries in the Colombian region of Casanare. According to the authors, oil exploration positively influenced public investment, job creation, and tax revenue in the region. Nevertheless, the discovery also made the area dependent and vulnerable to potential crises and negative external shocks.

Beyond the macroeconomic effects and theories related to the existence of natural resources and their relationship to development, it is essential to centrally understand the impact of oil on the Colombian economy. Doing so will allow for an empirical analysis of the impact of international price fluctuations on the country's economy, the primary objective of this article.

According to Fernández and Villar (2014), South American countries have benefited from significant natural resource discoveries in recent decades. Due to globalization and access to international markets, the discovery of these commodities positively impacted the economic growth of these countries. From the perspective of the Colombian economy, Melo-Becerra et al. (2016) note that oil activity has expanded since the 1980s and increased further in the 1990s. Between 2004 and 2014, Colombia experienced a period of high prices and export volumes, resulting in economic gains for the country. However, following the price decline in 2014, Colombia faced macroeconomic imbalances. According to the author, falling oil prices particularly impacted fiscal deficit and public debt variables, real exchange rates, the balance of payments, investment, and private consumption.

Llanes et al. (2018) also discuss the oil boom in the Colombian economy during the 2008–2016 period. During this period, total public investment in the country increased compared to other periods, with the transport and health sectors benefiting the most. The oil sector is capital-intensive and thus has limited effects on labor markets, but it significantly impacts the economy as a whole.

Given the importance of oil to the Colombian economy, this topic is analyzed from various perspectives. In particular, the effects of international oil price fluctuations on the Colombian economy are studied in different ways, such as the impact on public investment and the role of institutions (Bonet-Morón, Valbuena, and Llanes, 2020), oil shock impacts on Colombia's economic sectors (Otero, 2020), macroeconomic implications (Toro et al., 2016) and the role played by royalties (Bonet et al., 2023).

Methodologically, the aforementioned studies, except Bonet et al. (2023) employ econometric models to measure the magnitude of oil shocks and their impacts from different perspectives within the Colombian economy. These models include Difference-in-Differences (Bonet-Morón et al., 2020) and VAR (Otero, 2020). This study aims to contribute to this growing body of literature by adopting a computable general equilibrium (CGE) model. By treating oil price shocks as exogenous, this approach enables a systemic analysis of the country's economy in a post-shock scenario, assessing sectoral and macroeconomic impacts.

In macroeconomic terms, the results presented by Bonet-Morón et al. (2020) indicate that an increase in international oil prices has a positive effect on the country's investment levels. Meanwhile, Garavito, Lopez, and Montes (2016) show that negative oil price shocks (a price decrease) have a detrimental effect on the Colombian economy. Both findings align with the macroeconomic results obtained in this study, including positive impacts on GDP, investment, government spending, and terms of trade.

The findings from Otero (2020) also suggest a positive effect of oil price shocks on Colombia's economy in macroeconomic terms. In his sectoral analysis, the sectors most directly impacted by price shocks were manufacturing, industry, electricity, gas, and land transportation,

all of which are closely tied to oil production in the country. These results are consistent with the findings of this study, particularly in the gas distribution and transportation sectors, which, according to the CGE model, are significantly affected, primarily through price variations.

Beyond the aforementioned texts, the topic is also addressed from the perspective of computable general equilibrium, as adopted in this study. The dynamics of international oil prices are also discussed by some authors who aim to capture the impact of price increases in certain countries using computable general equilibrium (CGE) models (Fan et al., 2007; Timilsina, 2015). Fan et al. (2007) address this issue in the Chinese context, and Timilsina (2015) analyses it for a set of countries using the GTAP model.⁶ These studies point to a reduction in GDP (Fan et al., 2007; Timilsina, 2015) and in import and export flows (Fan et al., 2007) as international oil prices increase.

Beyond the scope of this study, some articles address the energy theme from other perspectives. Ribeiro et al. (2024) examine the productive sectors that could benefit from long-term economic growth in the state of Sergipe, Brazil. The authors point out that, although the oil and gas sector contribute to the growth of the state's GDP in the long run, it has negative effects concerning regional disparities. Meanwhile, Otsuka (2024) discusses the impact of exogenous shocks on electricity prices in Japan due to natural disasters and changes in electricity demand from households and industries.

3. Methodology and database

To calibrate the computable general equilibrium (CGE) model, among other data, it is necessary to define the input–output matrix, the basis of the model. To do so, the matrix provided by the National Administrative Department of Statistics (DANE), a Colombian government agency, will be used. This matrix was originally defined in Product X Sector format and encompasses 392 products and 60 sectors of the national economy. The model is calibrated for the year 2015 and for Colombian pesos (Haddad and Aroca, 2016). Based on this matrix, it becomes possible to conduct a descriptive analysis of trade flows in the country, with the aim of understanding the interconnection between different productive sectors of the economy.

Table 1 presents the destination of sales for each of the petroleum products. The destination structure varies significantly, reinforcing the characteristics of each product. When gasoline is blended with ethanol and liquefied petroleum gas, the main destination is the household. On the other hand, crude oil's main sales destination is the external sector, and gasoline for automobiles and kerosene are widely used for intermediate consumption.

Part of the crude oil production, intended for intermediate consumption, is used as an input by the oil refining sector, which was expected, given that crude commodity is the main input for refined products. In addition to the petroleum sector and the external market, crude oil is also used by the gas production and electricity generation sectors.

Furthermore, when analyzing the intermediate flows of refined products, especially gasoline for automobiles and kerosene, the land transportation sector and the air transportation sector are identified as the main demanders. This reflects the importance of fuels for the operation of the national mobility network.

To consolidate the main links in the oil production chain, a hypothetical extraction exercise was conducted. This simulation aims to

⁶ The model includes the following spatial disaggregation: Sub-Saharan Africa, Germany, Latin America and the Caribbean, Argentina, Australia and New Zealand, Brazil, Canada, China, Spain, United States, Central and Eastern Europe, Western Europe, France, India, Indonesia, Italy, Japan, Malaysia, Mexico, Middle East and North Africa, United Kingdom, Rest of Africa, Rest of Latin America and the Caribbean, Rest of East and Central Asia, Rest of Europe, Russia, and Thailand.

Table 1

Sales destination of petroleum products.

Product	Total Demand	Intermediate Consumption	Export	Household, Government, Investment
Crude Oil	100 %	19.80 %	81.10 %	0.10 %
Gasoline for automobile	100 %	87.90 %	11 %	0.10 %
Gasoline mixed with Ethanol	100 %	43 %	6 %	52.60 %
Kerosene	100 %	79.10 %	20 %	0.90 %
Petroleum Gas	100 %	42.30 %	5 %	49.40 %

Source: Prepared by the authors based on the Colombian Input-Output (I-O) Matrix (Dane, 2015).

Table 2

Main linkages in the Oil Production Chain.

Sectors	
Land Transports	\$44
Gas Production and Distribution	\$35
Manufacture of basic chemical substances	\$25
Electricity Generation	\$34
Wholesale Trade	\$42
Financial Activities	\$51
Scientific and technical activities	\$53
Administrative and support service activities	\$54

Source: Prepared by the author based on the I-O Matrix for Colombia.

quantify the reduction in production of an economy due to the total or partial shutdown of one or more sectors, thus measuring the linkage of that sector in the analyzed economy (Miller and Blair, 2022). By performing the hypothetical extraction of the crude oil extraction and petroleum refining sectors, it was possible to identify the main links of the oil production chain in the Colombian economy, i.e., those sectors that are most related to petroleum. Given the systemic interconnection with oil production, these sectors tend to suffer the greatest impacts in scenarios of exogenous shocks to the country's economy. Table 2 presents the oil production chain in Colombia, illustrating the sectors that experience the largest production losses following a hypothetical extraction in oil-producing sectors. According to the input-output simulation, these are the sectors most affected when petroleum production is interrupted. This table provides insights into the sectors most tightly interconnected with the petroleum industry, particularly those with the strongest linkages in terms of both forward and backward linkages. By mapping these connections, it highlights the sectors most vulnerable to disruptions in oil production and emphasizes their critical role within the broader economic structure.

3.1. The computable general equilibrium model

Using the computable general equilibrium (CGE) model, the impact of increasing international oil prices on the Colombian economy is measured. In general, CGE models use real economic data to predict the impacts of external factors on an economy, such as tariff changes and policy alterations, among other potential shocks that may affect one or more regions. (Dixon and Jorgenson, 2012) noted that CGE models aim to demonstrate real-world policy issues and generate insights into the effects of potential shocks on areas such as trade, taxation, public spending, social security, demographics, labor markets, and other areas, allowing for the quantification of impacts on local industries and socioeconomic groups.

This article uses the Orani_G⁷ model calibrated for Colombia (Haddad and Aroca, 2016). The model is calibrated based on the Colombian Input–Output Matrix for the year 2015 in its original format, i.e., Product x Sector. According to Dixon (1982), Orani_G enables this formulation because, in real scenarios, industries produce more than one product, making this approach feasible.

Horridge (2006) presented the production structure of the Orani_G model.⁸ As mentioned, each sector can produce one or more products using domestic or imported inputs. The primary production factors in the model are land, labor, and capital. The author indicated that in the model, the equilibrium of primary factors is obtained through a composite constant elasticity of substitution (CES) function, where the three primary factors present in the model interact. The intermediate goods of the model are also defined by CES functions, with proportions divided between imported and domestic inputs. Thus, the baskets of goods that compose the economy's inputs, in conjunction with the primary factors under a Leontief function, will result in a level of economic activity. Finally, goods are produced following a constant elasticity of transformation (CET) function, and these goods can be destined for the domestic market or for export. The household consumption structure is also developed at the optimization level. Households choose between domestic or imported goods through a CES function. The level of utility of households is determined through a Klein–Rubin function, where users allocate their resources to luxury or subsistence goods, controlled by Armington elasticity.

For the implementation of the simulation exercise, long-term closure was used. In this scenario, according to Horridge (2006), capital stocks become endogenous and adjust in the model, while the rates of return are fixed. Therefore, an open capital market is assumed. The employment rate is also fixed, while adjustments to real wages are permitted. Finally, from the expenditure perspective, GDP is adjusted by household and government consumption, and the ratio of the trade balance to GDP is fixed. Aggregate investment will follow the capital stock. The closure details, including exogenous and endogenous variables, are described in the Appendix A.

4. Simulation exercise

To simulate the impacts of increasing international oil prices on the Colombian economy, a shock will be simultaneously applied to the exogenous variable " f_{4pc} ," representing a variation in the prices of exports demand for a specific commodity, and to the variable " $pf0cif$," which pertains to a variation in the prices of imports. This approach allows for a comprehensive measurement of the impacts of increases in international oil prices.

In general terms, modifying world oil prices changes the behavior of export demand, which, given the closure of the model, is guided by the ratio between domestic prices and world prices. The external demand for Colombian products, c , x_{4c} , is a function of the average price in foreign currency ($p_{4c} - \varphi$) and displacement variables (f_{4xc} , f_{4pc}).

$$x_{4c} \cdot f_{4xc} = \sigma_{Ex, c} (p_{4c} - \varphi - f_{4pc})$$

where φ is the percentage change in the exchange rate (exogenous model currency) and $\sigma_{Ex, c}$ is a constant price elasticity parameter of demand. Therefore, those products that experience a change in the average prices in foreign currency ($p_{4c} - \varphi$), greater than the variation in world prices ($f_{4pc} = 1\%$), will experience a reduction in their export

demand, and vice versa. It is worth noting that export demands are decreasing functions of ($p_{4c} - \varphi$). Domestic prices are endogenous and adjusted according to the simulations.

In summary, the immediate consequences of a variation in world prices (f_{4pc}) can be described as follows:

- Impact on the export demand curve – (x_{4c}). The magnitude of the variation in export volume also directly depends on the prices of the exported products ($p_{4c} - \varphi$) and the elasticity of export demand ($\sigma_{Ex, c}$); the price of exported products is affected by internal production costs, which depend on the relative price of factors and production inputs.
- The variation in export volume (x_{4c}) has a direct impact on the balance between supply and demand in the nonmargin goods market.
- Adjustment in the nonmargin goods market from the supply side, meaning the shift of the export demand curve, can be understood as an increase in the "preference" to offer goods outside the country rather than domestically. However, this shift in sales allocation may be limited by supply constraints, i.e., production capacity, and the increase in production costs (in the short-term closure of the model, investment and capital stock are fixed). Therefore, adjustments may occur in consumption, investment (only in the long term), and government spending (domestic absorption).

The sequence of the shock in the variable f_{4p} can also be understood through the following flowchart.

From the perspective of imports, an effect on the volume imported by the country is expected, which consequently will affect the domestic supply and demand for these external goods. Under such circumstances, the impact occurs directly on the basic prices of imports, as presented by the equation below.

$$p_0 = pf0cif + \varphi + t_{imp}$$

Given the adopted closure, the exchange rate (φ) and import tariffs are defined as exogenous; therefore, the shock directly affects the basic prices of imported goods. The shock will be applied to all petroleum-

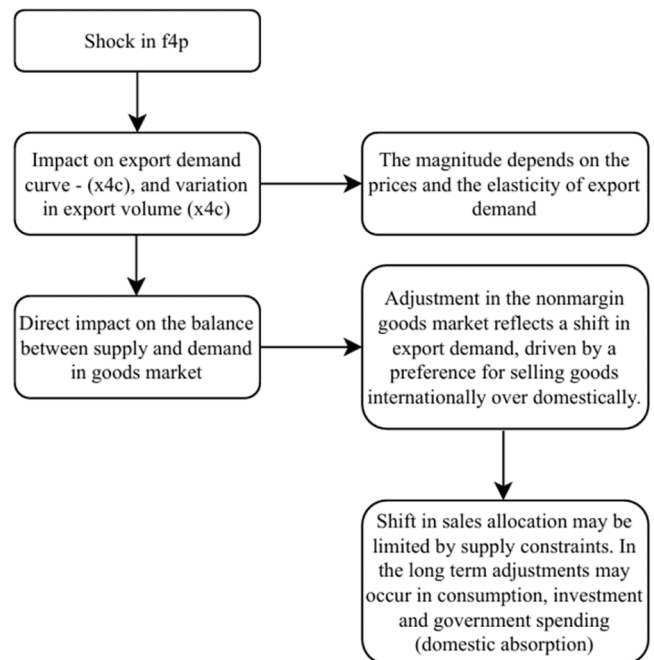


Fig. 2. F4p shock sequence.
Source: Prepared by the authors

⁷ Orani_G was initially formulated as a disaggregated model for the Australian economy in 1977. It has been used since then to analyze the effects on industries and the labor market and has been used in regional analyses based on the simulation of variations in tariffs, world prices, rate interest rates, application of subsidies, and other factors (Dixon and Jorgenson, 2012).

⁸ For more details on the nested structure of the model, see Horridge (2006).

derived products present in the Colombian economy. These include crude oil, gasoline for automobiles, gasoline blended with ethanol, aviation kerosene, and liquefied petroleum gas. Thus, the objective of simulating the increase in prices of related products stems from their connection with crude oil, assuming that an increase in international prices is also reflected in their petroleum-intensive derivatives.

For the proposed simulation, the aim is to apply an ad hoc shock of 1 % to the world prices of the commodity, thus capturing the implicit elasticity of prices in the Colombian economy. The main objective is to identify the movement and signs of variables rather than to make predictions regarding future oil prices.

Due to the computable general equilibrium model using various elasticities in its equations, it is valid to present the values of the Armington and Export elasticities for petroleum products, as these values will impact the results obtained.

The Armington elasticity is included in the equations for household consumption, investment, and intermediate consumption. This elasticity affects the choice of economic agents between domestically produced and imported commodities, depending on the price. On the other hand, export elasticity is included in the export demand equation. Products with higher absolute values tend to be more in demand externally. The elasticities used in the model were estimated by [Haddad and Aroca \(2016\)](#) and are shown in [Table 3](#).

4.1. Results

In macroeconomic terms, an increase in international oil prices can result in a positive percentage change in GDP, driven by a positive percentage change in household consumption, investment levels, and government spending. [Table 4](#) presents the macroeconomic results after the simulation.

The positive variation in GDP is driven by the increase in government spending, which shifts positively. Investment and household consumption also show positive variations. The positive variation in investments is partly a reflection of the need to increase production to meet the new demand in the national economy. Despite the increase in petroleum-derived product prices and, consequently, the higher cost of goods such as gasoline, there was an increase in household consumption. This movement can be explained by the fact that the most affected goods do not correspond to a significant portion of the household consumption basket in the model. Due to the increase in investments and productive demand, there was an increase in workers' wages, at a level higher than that identified in the price index, resulting in a real increase in wages and consumption. The macroeconomic results following a positive oil price shock, which include positive impacts on GDP, investment, government spending, and terms of trade, are consistent with the literature and are also documented in the works of [Bonet-Morón et al. \(2020\)](#); [Garavito, Lopez, and Montes \(2016\)](#).

The percentage variation in imports exceeds that in exports, indicating a potential trade deficit in the country in the long term, especially due to the reduction in the volume exported by the country. The terms of trade in the Colombian economy also showed positive variation, indicating that domestic prices are higher than international prices. This may reflect a loss of international competitiveness in terms of exports.

Table 3
Elasticity parameters.

Products		Armington Elast.	Exportation Elast.
Crude Oil	P57	2,6	−0,420
Gasoline for automobile	P157	1,05	−0,648
Gasoline mixed with Ethanol	P158	0	−0,648
Kerosene	P159	1,05	−0,648
Petroleum Gas	P162	1,05	−0,648

Source: Prepared by the author based on the model calibrated for Colombia ([Haddad and Aroca, 2016](#))

Table 4
Macroeconomic indicators.

Macroeconomic Indicators	Var %
GDP	2,27
Trade Balance	−2,27
Exported Volume	−0,45
Imported Volume	1,10
Household Consumption	1,27
Investment	0,97
Government Spending	2,03
Real Salary	1,50
Consumer Price Index	1,47
Terms of Trade	0,97

Source: Prepared by the author based on the results.

On the other hand, because external products are cheaper, an increase in imports is expected. The decline in exports is a direct result of the increase in the price of crude oil, which represents the country's main export product. Therefore, the loss of external competitiveness in this product directly impacts the Colombian trade balance. When an exogenous shock is established in oil prices, it becomes necessary to assess the pass-through of this increase to Colombian products. Such analysis will be described through the percentage variations in basic prices described on [Table 5](#).

After the shock, in the long term, there was a positive variation in the basic prices of almost all products traded in the country. The variation for those products that directly received the shock in prices is noteworthy. For blended gasoline, liquefied petroleum gas, and crude oil, the variation exceeded the value of the shock, 1 %. For such products, domestically produced goods experienced an even greater increase in price than world prices. As a result, these products tend to lose international competitiveness in the long term, as they are relatively more expensive than external prices are. In this way, especially regarding crude oil, there was a reduction in the international participation in this Colombian product, which can directly reflect the country's export decline, as this product is the main item in Colombia's export basket.

In addition to the price variation, it is possible to assess the change in intermediate costs of productive sectors in the country. This occurs because sectors acquire their inputs and trade their productions with other productive sectors in addition to the elements of final demand. The multiproduction nature of the Input-Output Matrix used in this simulation is emphasized; therefore, each productive sector can produce one or more goods. [Table 6](#) presents the variation in intermediate costs incurred by productive sectors after the shock in the long term.

Due to the overall price variation in the economy, there is a rise in intermediate costs for all sectors belonging to the input–output matrix. The sector of gas production and distribution stands out due to its direct productive interconnection with the petroleum sector. This connection arises because its main input, coal, is predominantly produced by the crude oil extraction sector. Therefore, a variation in the prices of the crude commodity indirectly affects the prices of the coal input, thereby increasing the costs of the gas production and distribution sector.

For the land transport and air transport sectors, the use of refined petroleum products, such as gasoline and kerosene, stands out as their main inputs in their production processes. Therefore, shocks in these commodities directly impact the costs of these transport sectors.

Table 5
Basic prices.

Products	p0	Var %
Gasoline mixed with Ethanol	P158	1,69
Petroleum Gas	P162	1,63
Crude Oil	P57	1,12
Gasoline for automobile	P157	0,95
Kerosene	P159	0,69

Source: Prepared by the author based on the results.

Table 6
Intermediate input price index – long term.

Sector	p1mat	Var %
Gas products and Distribution	S35	1,55
Petroleum Refining Products	S24	1,21
Land Transports	S44	1,18
Crude Oil Extraction	S6	1,06
Air Transport	S46	1,02

Source: Prepared by the author based on the results (2023).

In this case, the sectoral impact is primarily confined to the oil production chain, with these sectors being the most affected in terms of increased production costs resulting from oil price fluctuations. These findings are consistent with previous studies, as highlighted by Otero (2020) and the results confirm the hypothesis raised by Jiménez (2010) that the increase in oil prices can affect production across the country and have direct and indirect impacts on a country's economic growth and other macroeconomic variables. Finally, both petroleum sectors also experienced an increase in their intermediate costs, which is explained by the use of their products as productive inputs. Moreover, the main input of the refined petroleum products sector is crude oil, which incurred a price increase after the shock.

In addition to the price and intermediate cost variations, the country's production is also impacted both at the national level and by imports, which also supply the domestic market. The results of the percentage variation in supply are described in Table 7.

When assessing the impacts of the shock on production, it is observed that all petroleum products had their production boosted despite the price variation, thus increasing the national supply of these goods. From the perspective of imports, positive values are highlighted for all goods, indicating that the domestic supply is also supplemented by imported products after the shock. The increase in imports may be the result, in part, of the superior variation in domestic prices compared to imported prices. The decomposition of supply, aimed at identifying the reasons for the production variation, is presented in Table 8.

Analysis of the Destination of the Positive Variation in the Production of Petroleum Products reveals that the primary boost for these products is the domestic market. Examining domestic participation, it is evident that only kerosene shows significant positive variation. This is because its domestic prices are lower than its imported prices, inducing a process of import substitution for this product.

The same trend is observed for exports, where only kerosene and gasoline for automobiles have a positive export effect, which is also explained by relative prices. Despite being Colombia's main export product, crude oil has a negative export effect, indicating that an increase in its prices relative to world prices reduces its competitiveness. Given the importance of crude oil in Colombia's trade balance, this movement could be one of the catalysts for the increase in the country's trade deficit in long-term simulations.

To capture the systemic effects of the shock, Table 9 presents the percentage variation in intersectoral intermediate flows. For this purpose, sectoral aggregation was performed.

It is evident that the effects spread more heterogeneously across the productive sectors in Colombia. For example, the sectors encompassing petroleum products (extractive and heavy manufacturing) are not

among the sectors that have experienced the most variation in their intermediate flows.

Due to the overall increase in prices throughout the economy, including to a greater extent for other products than for petroleum products, inputs for all sectors have become more expensive. Consequently, there has been an increase in intermediate flows in monetary terms. In agriculture, the livestock and fishing sectors exhibited notable increases in their flows, which can be explained by increased household consumption. The aggregation of extractive sectors, on the other hand, includes the only sector that individually experienced a reduction in its intermediate flows, the disaggregated sector of activities supporting extraction.

The transportation, trade, and communications sectors, on the other hand, showed positive variation in sectors interconnected with the petroleum production chain, such as trade and wholesale, land transportation, and air transportation, demonstrating that the variation in petroleum commodity prices also impacts those sectors linked to petroleum production.

4.2. Decomposition of shocks

When employing a computable general equilibrium model and defining more than one simultaneous shock, each modification in the endogenous variables will depend on the shocks applied. According to Harrison and Horridge (1999), these distinct shocks can be decomposed to measure the contribution of each shock to the variations after the simulation. The decomposition of the macroeconomic indicators is presented in Table 10.

The positive percentage variation in Colombian GDP (1.69 %) in the long term is mostly linked to the variation in the price of export demand (2.28 %). Regarding export volume, a reduction is observed due to the combined shocks. However, once again, crude oil is responsible for most of this reduction. This is because, after the simulation shock, in the long term, Colombian crude oil became relatively more expensive than international oil was, leading to a loss of competitiveness and a consequent reduction in exports of this commodity. Since it is the country's main export product, this decrease in volume directly impacts Colombian total exports.

From the perspective of import volume, it is noticeable that the variation in the price of exported goods (f4p) had a greater contribution. This movement aligns with the variation in export volume, indicating a loss of competitiveness of the product in relative terms, leading to an increase in imports of this commodity, which traditionally has a trade surplus. On the other hand, the variation in the price of imports reflects a reduction in import volume due to the increased cost of these goods.

Regarding both investment and government spending variations, it is noted that the shock in world prices positively drives both variables. This occurs, in part, due to the need for greater investment due to price increases. The variation in import prices has a negative effect on these variables due to the reduction in the prices of some imported goods, which hinders investment.

Table 11 indicates the results of the decomposition of shocks on basic prices.

When analyzing the variation in the basic prices of petroleum products in the long term, it is observed that those in which the variation

Table 7
Supply and import of goods.

Products	Production	Var%	Products	Importation	Var%
Gasoline mixed with ethanol	P158	0,872	Gasoline mixed with ethanol	P158	0,169
Petroleum Gas	P162	0,849	Petroleum Gas	P162	1,494
Gasoline for automobiles	P157	0,509	Gasoline for automobiles	P157	0,621
Kerosene	P159	0,381	Kerosene	P159	0,238
Crude Oil	P57	0,077	Crude Oil	P57	0,931

Source: Prepared by the author based on the results (2023).

Table 8

The decomposition of supply.

Sectors	Decomposition	Local Market	Domestic Participation	Exportation	Total
Gasoline mixed with ethanol	P158	0,904	0,000	−0,032	0,872
Petroleum Gas	P162	0,900	−0,029	−0,022	0,849
Gasoline for automobiles	P157	0,483	0,018	0,007	0,509
Kerosene	P159	0,173	0,116	0,091	0,381
Crude Oil	P57	0,118	−0,001	−0,003	0,077

Source: Prepared by the author based on the results (2023).

Table 9

Intersectoral intermediate flows.

Sectors	Var %
Agriculture	1,705
Extractive Sector	1,048
Light Manufacturing	0,896
Heavy Manufacturing	1,224
Utilities and Construction	1,405
Transportation, Trade and Communications	1,896
Other Services	1,981

Source: Prepared by the author based on the results (2023).

Table 10

Decomposition of the macroeconomic indicators.

Macroeconomic Indicators	Var %	F4p	Pf0Cif
GDP	1,69	2,28	−0,59
Exported Volume	−0,45	−0,62	0,17
Import Volume	1,04	1,44	−0,39
Employment Variation	0	0	0
Consumer Price Index	1,48	1,99	−0,51
Household Consumption	1,27	1,75	−0,48
Investment	0,97	1,30	−0,33
Government Spending	2,03	2,73	−0,69
Terms of Trade	0,97	1,32	−0,35

Source: Prepared by the author based on the results (2023).

exceeded the value of the shock (1 %) are the ones that were most impacted by the variation in world prices of the commodity. This occurs because, for these products, the transmission of the shock in f4p made such goods more expensive than in the external market. Therefore, the importation of these goods becomes viable and reduces the variation in basic prices.

The impacts are closely linked to the variation in relative prices; thus, products that experience a transmission greater than the value of the shock become less attractive domestically and may be substituted by imported products. The same logic applies to products whose main destination is exports, such as crude oil.

4.3. Sensitivity analysis

In a computable general equilibrium model, several elasticities are used, constituting the parameters of the model. These parameters can be established through econometric estimates or assumptions based on the behavior of each evaluated economy. Therefore, with the aim of testing the model's robustness, a sensitivity analysis is proposed.

In the systematic sensitivity analysis of this work, the Gaussian quadrature methodology proposed by [Deyust and Preckel \(1997\)](#) was used, as described in some works, such as [Domingues et al. \(2003\)](#), [Perobelli \(2004\)](#), [Burfisher \(2021\)](#). The computable general equilibrium model is treated as a numerical integration problem where the solution of the model can be obtained simultaneously with the results of the endogenous variables and their first two moments (mean and standard deviation) given a variation in the parameters.

The robustness test of the parameters showed a variation of $\pm 10\%$ in export elasticities and Armington elasticity. This variation seeks to

Table 11

Decomposition of basic prices.

p0	Var %	F4p	Pf0Cif
Crude Oil	1,12	1,41	−0,28
Gasoline for automobile	0,955	0,504	0,411
Gasoline mixed with ethanol	1,685	1,989	−0,303
Kerosene	0,698	0,213	0,485
Petroleum Gas	1,639	1,881	−0,241

Source: Prepared by the author based on the results (2023).

identify the movement of the variables given a modification in the elasticities used to calibrate the model. Thus, it is expected that there is no variation in the signs, demonstrating that the parameters are robust.

5. Final considerations

In general terms, after simulating the change in global oil prices, it was possible to assess the recurring impacts of this shock on the Colombian economy. In macroeconomic terms, there is an increase in GDP, mainly driven by higher investment and government spending. Overall, there is a positive variation in real wages after the shock, which can also be reflected in increased household consumption. However, the systemic results for the Colombian economy in the long run also point to a generalized positive variation in the basic prices of the economy.

The positive variation in prices does not negatively affect production; there is an increase in the supply of goods in the country, driven in part by increased real household consumption and intermediate flows. A higher demand for products is reflected in the increased intermediate flows of sectors in the country due to a greater need for intermediate inputs.

Historical analysis of Colombia's international market behavior revealed that in times of high oil prices, the Colombian economy tends to perform positively. According to the Colombian Input–Output Matrix, in the model's calibration year, crude oil accounts for 32 % of the country's total exports. However, during the adoption of the closure, the increase in the price of the Colombian product had a negative impact on its exports.

According to the simulation shock, the main impact on the Colombian economy is the transmission of this shock to the basic prices of the economy. From the perspective of the trade balance, imports increase more than exports, widening the country's trade deficit. When analyzing

Table 12

Sensitivity test – export elasticity.

Macroeconomic Indicators	Long Term		
	Simulation %	Mean	Standard Deviation
GDP	2,2711	2,3682	0,4183
Export Volume	−0,4578	−0,4780	0,8666
Import Volume	1,0963	1,1406	0,1913
Employment Variation	0	0	0
Real Household Consumption	1,2708	1,3248	0,2346
Consumer Price Index	1,4858	1,5487	0,2172
Terms of Trade	0,9763	1,0181	0,1801

Source: Prepared by the author based on the results (2023).

Table 13
Sensitivity test–Armington elasticity test.

Macroeconomic Indicators	Long Term		
	Simulation %	Mean	Standard Deviation
GDP	2,2711	2,2903	0,2020
Export Volume	−0,4578	−0,4626	0,5015
Import Volume	1,0963	1,1044	0,0855
Employment Variation	0	0	0
Real Household Consumption	1,2708	1,2814	0,1129
Consumer Price Index	1,4858	1,4983	0,1311
Terms of Trade	0,9763	0,9845	0,0855

Source: Prepared by the author based on the results (2023).

the effects on crude oil, a decrease in external demand for this product was observed due to the high price relative to the global variation. Therefore, there is a loss of international competitiveness for this product, which may indicate a need for diversification of the export basket in the long term in scenarios of high commodity prices.

In the long term, in systemic terms, when evaluating the variation in total intermediate and production flows, the greatest variations are concentrated in sectors that are part of the oil production chain, such as gas production and distribution, land transport and wholesale trade. This analysis corroborates the conclusion that the country's oil production does not have a high level of interconnection, resulting in a low spread of results to other sectors. This fragmentation of the oil production process can impact the country's growth due to the sector's low connection with the economy as a whole. In addition, there is high participation of the commodity in national exports, which demonstrates the need for diversification of the export basket due to the existing dependence on crude oil. In the long-term scenario, where commodity prices vary to a greater extent than international prices, there is a loss of external competitiveness for the country. It is therefore suggested that economies with a low degree of diversification tend to become more vulnerable to commodity price volatility, as the loss of competitiveness in these products can lead to an increase in the trade deficit. This was also observed for imported products, which increased their entry into the country after the shock in the long term, indicating a potential external dependence on other products as well.

This study, therefore, aimed to identify the main systemic impacts of oil price volatility, as well as the sectoral effects of potential price shocks. The findings highlight that economies with limited diversification and heavy reliance on commodities and primary products are more vulnerable during periods of scarcity or crises driven by price fluctuations. Countries like Colombia, where oil is a significant source of revenue and a key driver of exports, should prioritize diversification in both their productive and export sectors to strengthen their resilience against external shocks. It is important to emphasize that, as a price-taking nation with no direct influence over oil pricing, Colombia's economy must be particularly prepared to withstand the adverse effects of the inherent and ongoing fluctuations in oil prices.

The evidence presented demonstrates that oil shocks directly affect sectors closely linked to oil, particularly petroleum refining products, land transportation, air transportation, and electricity generation. These findings suggest that mitigating the adverse effects of oil shocks may require diversifying the country's energy model to reduce internal reliance on petroleum-based fuels. Public policies promoting the production and adoption of biofuels such as: ethanol, biodiesel, and sustainable aviation fuel could help diversify the transportation sector. In the electricity generation sector, the integration of photovoltaic modules could contribute to reducing the impacts of oil price fluctuations. It is necessary to put these agenda in perspective, since this kind of investment has long-term maturity.

The model employed in this study aimed to assess the sectoral impacts of oil price fluctuations in Colombia, without considering regional variations in the effects of the shock. Since the distribution of oil wells

across the country is neither random nor homogeneous, it can be assumed that the impacts of the simulated shock will differ across Colombia's departments. The exclusion of regional impacts represents a limitation of this study, one that could be addressed in future research by implementing an inter-regional computable general equilibrium model. Such an approach would capture the diverse inter-regional trade flows between the country's departments, offering a more granular understanding of the economic consequences of oil price fluctuations.

CRedit authorship contribution statement

Perobelli Fernando Salgueiro: Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization. **Salgueiro Guilherme Perobelli:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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